

LeetCode 695: Max Area of Island

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Problem Overview

Max Area of Island Example

Solution 1: Breadth-First Search (BFS)

```

class Solution {
public:
    int maxAreaOfIsland(vector<vector<int>>& grid) {
        int n=grid.size();
        int m=grid[0].size();
        vector<pair<int,int>> dir={ {-1,0},{1,0},{0,-1},{0,1} };
        queue<pair<int,int>> q;
        int maxu=0;
        int ans=0;

        for(int i=0;i<n;i++){
            for(int j=0;j<m;j++){

                if(grid[i][j]==1){
                    ans=0;    //reset the counting

                    grid[i][j]=0; //to mark it as visited and avoid duplicate counting
                    q.push({i,j}); //push to queue to start counting
                    while(!q.empty()){

                        int r=q.front().first;
                        int c=q.front().second;
                        q.pop();
                        ans++; //increase length of island

                        for(auto it:dir){
                            int nr = r + it.first;
                            int nc = c + it.second;

                            if (nr >= 0 && nr < n && nc >= 0 && nc < m){
                                if(grid[nr][nc]==1){
                                    grid[nr][nc]=0; //push krne se pehle hamesha
                                    q.push({nr,nc});
                                }
                            }
                        }
                    } //1 island yaha htm hua

                    maxu=max(maxu,ans);
                }
            }
        }

        return maxu;
    }
};

```

```
}  
};
```

Solution 2: Depth-First Search (DFS)

```

class Solution {
public:

    vector<pair<int,int>> dir={ {-1,0},{1,0},{0,-1},{0,1} };

    void dfs(int r,int c,vector<vector<int>>& grid,int &cnt){

        grid[r][c]=0;
        cnt++;

        int n=grid.size();
        int m=grid[0].size();

        for(auto it:dir){

            int nr=r+it.first;
            int nc=c+it.second;

            if(nr>=0 && nr<n && nc>=0 && nc<m){

                if(grid[nr][nc]==1){

                    dfs(nr,nc,grid,cnt);

                }

            }

        }

    }

    int maxAreaOfIsland(vector<vector<int>>& grid) {

        int n=grid.size();
        int m=grid[0].size();

        int maxu=0;
        int cnt;

        for(int i=0;i<n;i++){

            for(int j=0;j<m;j++){

```

```
        if(grid[i][j]==1){  
            cnt=0;  
            dfs(i,j,grid,cnt);  
            maxu=max(maxu,cnt);  
        }  
    }  
}  
return maxu;  
}  
};
```